Executive Summary – Customer Churn Analysis

1. Project Overview

Customer churn is a major business challenge for subscription-based companies because losing existing customers can reduce recurring revenue and increase the need to acquire new customers. The purpose of this project is to analyze customer churn, understand customer behavior, and identify the customer segments and service-related factors associated with customer attrition.

In this project, I used Python, Pandas, NumPy, Matplotlib, and Seaborn to perform data cleaning, exploratory data analysis (EDA), statistical analysis, and data visualization. I created multiple charts to examine churn patterns across customer demographics, contract types, payment methods, tenure, and subscribed services.

The analysis is based on **7,043 customer records and 21 columns**, including customer information, subscription details, billing information, service preferences, and churn status.

2. Data Preparation and Quality Checks

Before conducting the analysis, I performed several data preparation steps to improve data quality and ensure the dataset was suitable for visualization.

- **Dataset inspection:** Examined the dataset structure, column names, data types, and number of records using Pandas.
- **Missing-value handling:** Identified blank values in the `TotalCharges` column and replaced blank entries with zero, particularly for customers with zero tenure.
- **Data type conversion:** Converted `TotalCharges` into a numeric format to support statistical analysis.
- **Missing-value validation:** Checked missing values across all 21 columns. The notebook reports zero missing values after preprocessing.
- **Duplicate validation:** Checked for duplicate rows and duplicate customer IDs. Both checks returned zero duplicates.
- **Categorical data preparation:** Converted the `SeniorCitizen` variable from numerical values (0 and 1) into readable categories (No and Yes).

These steps established a consistent dataset for the subsequent analysis. However, treating blank total charges as zero is an assumption that should be validated against the original customer records.

3. Overall Customer Churn Analysis

The overall churn analysis establishes the baseline for understanding customer retention.

Metric	Customers	Percentage
Total customers	7,043	100.00%
Customers who churned	1,869	26.54%
Customers who remained	5,174	73.46%

The results indicate that approximately **26.54% of customers have churned**, while 73.46% remain with the company. In other words, approximately 27 out of every 100 customers have discontinued the service.

This finding is important because churn affects the stability of recurring revenue and increases the pressure on the business to acquire replacement customers. The overall churn rate also provides a baseline against which individual customer segments can be compared.

Business implication: The company should establish a regular churn-monitoring process and prioritize customer segments whose churn rates are substantially higher than the overall rate of 26.54%.

4. Customer Demographics Analysis

4.1 Gender Analysis

The gender distribution and gender-based churn charts help compare customer counts and churn behavior between male and female customers.

The analysis considers two important questions:

- How are customers distributed across gender categories?
- Does the percentage of customers who churn differ between male and female customers?

The notebook includes both count-based and stacked bar charts. These visualizations make it possible to compare churned and retained customers across the two groups.

Customer counts alone should not be interpreted as evidence that one gender is more likely to churn. The churn rate within each gender should be calculated separately using the total number of customers in that gender category as the denominator.

Business implication: Gender can be monitored as a customer segmentation variable, but retention decisions should be based on meaningful differences in churn rates rather than raw customer counts alone.

4.2 Senior Citizen Analysis

The notebook converts the senior citizen indicator into Yes/No categories and explores customer demographics using count plots.

This analysis helps the business compare the distribution of senior and non-senior customers. It can also support further investigation into whether age-related customer segments experience different retention patterns.

To make the analysis more informative, churn rates should be compared separately for senior and non-senior customers.

Business implication: If a meaningful difference in churn rates is identified, the company can investigate whether customer support, service preferences, or billing experience differs between these groups. The current chart alone does not establish the cause of any difference.

5. Customer Tenure Analysis

Customer tenure represents the length of time a customer has remained with the company, measured in months. The notebook uses a histogram with churn status to examine the distribution of tenure across churned and retained customers.

This visualization is useful for identifying whether churn is concentrated among newer customers or distributed across different stages of the customer lifecycle.

The tenure analysis supports three areas of investigation:

- **Early-stage retention:** Examine whether customers with shorter tenure have a higher likelihood of leaving.
- **Long-term retention:** Determine whether customers with longer relationships are more likely to remain subscribed.
- **Customer onboarding:** Investigate whether early service issues, unclear billing, or unmet expectations contribute to early customer departures.

The descriptive statistics show an average tenure of approximately **32.37 months**, a median tenure of **29 months**, and a maximum tenure of 72 months.

These figures describe the overall customer base; they do not independently establish the churn rate at each tenure level.

Business implication: The company should consider targeted onboarding, early engagement, and proactive customer support, particularly if further analysis confirms that customers with shorter tenure have higher churn rates.

6. Contract Type Analysis

The contract analysis compares churned and retained customers across month-to-month, one-year, and two-year contracts.

Contract type	Churned customers	Share of all churned customers	Churn rate within contract type
Month-to-month	1,655	88.55%	42.71%
One year	166	8.88%	11.27%
Two years	48	2.57%	2.83%
Total	**1,869**	**100.00%**	**26.54% overall**

Percentages are calculated from the customer counts and category totals in the standard 7,043-record Telco Customer Churn dataset represented by this notebook.

The results show a substantial difference in churn behavior across contract types. Month-to-month customers account for approximately 88.55% of all churned customers, and their churn rate is approximately 42.71%.

By comparison, customers on one-year contracts have a churn rate of approximately 11.27%, while customers on two-year contracts have a churn rate of approximately 2.83%.

This indicates a strong association between contract type and customer retention. Customers with longer-term contracts are less likely to churn in this dataset.

Business implication: The company could test loyalty benefits, renewal incentives, and suitable long-term contract offers for month-to-month customers. Such offers should be evaluated for their effect on retention, customer satisfaction, and profitability rather than assuming that every customer will prefer a longer contract.

7. Payment Method Analysis

The payment method chart compares churned and retained customers across electronic checks, mailed checks, bank transfers, and credit card payments.

Payment method	Churned customers	Share of all churned customers	Churn rate within payment method
Electronic check	1,071	57.30%	45.29%
Mailed check	308	16.48%	19.11%
Bank transfer (automatic)	258	13.80%	16.71%

Credit card (automatic)	232	12.41%	15.25%
Total	**1,869**	**100.00%**	**26.54% overall**

The analysis identifies electronic check users as the largest churn group. They account for approximately 57.30% of all churned customers, and their category-specific churn rate is approximately 45.29%.

In comparison, customers using automatic bank transfers and automatic credit card payments have churn rates of approximately 16.71% and 15.25%, respectively.

This pattern suggests that payment method is a useful variable for further investigation. However, the chart does not prove that electronic checks directly cause customer churn. Other characteristics, including contract type and tenure, may also influence the observed relationship.

Business implication: The company should investigate the billing experience of electronic check users, including payment convenience, failed payments, and customer preferences. It could also test whether encouraging convenient automatic payment options improves retention.

8. Customer Services and Subscription Features

The notebook contains additional visualizations covering the following variables:

- Partner and dependent status
- Phone service and multiple lines
- Internet service type
- Online security and online backup
- Device protection and technical support
- Streaming TV and streaming movies
- Paperless billing

These charts provide a broader view of the customer base and help identify how subscriptions and service preferences are distributed.

The churn-by-category charts are especially useful because they allow the business to examine whether particular services or customer groups have different churn patterns.

For example, the company can investigate whether customers without technical support or online security experience different churn rates from customers who subscribe to those services. Similarly, paperless billing and internet service type can be explored as possible indicators of differences in customer behavior.

The analysis should compare the percentage of churned customers within each category rather than relying exclusively on total counts. Where multiple variables are associated with

churn, additional analysis is needed to determine whether the relationships remain after accounting for contract type, tenure, and monthly charges.

Business implication: Service-related findings can help the company develop targeted support and service-improvement initiatives. Any proposed intervention should be tested against measurable retention outcomes.

9. Financial and Billing Overview

The notebook's descriptive statistics provide additional context about customer billing.

Metric	Value
Average monthly charges	64.76
Median monthly charges	70.35
Average total charges	2,279.73
Median total charges	1,394.55
Average tenure	32.37 months
Median tenure	29 months

The currency is not explicitly specified in the notebook, so the monetary figures are reported without assigning a currency symbol.

Average monthly charges are approximately 64.76, while average total charges are approximately 2,279.73. The difference between the average and median total charges suggests that customer billing totals vary considerably across the dataset.

These statistics provide useful background for further analysis of the financial impact of churn. However, the current notebook does not quantify the revenue lost specifically from churned customers.

Recommended next step: Compare monthly charges and total charges for churned versus retained customers, and estimate the recurring revenue associated with customers at elevated risk of leaving.

10. Key Business Insights

The combined analysis highlights several important findings:

1. **Churn is a significant retention challenge.** Approximately 26.54% of customers have churned, providing a baseline for future retention initiatives.

2. **Contract type is a major area of concern.** Month-to-month customers have a churn rate of approximately 42.71%, compared with 2.83% for two-year contract customers.
3. **Electronic check users require further investigation.** Their churn rate is approximately 45.29%, substantially higher than the overall churn rate.
4. **Tenure provides useful customer lifecycle context.** The distribution can help identify whether early-stage customers require additional engagement and support.
5. **Customer demographics and service preferences provide further segmentation opportunities.** Gender, senior citizen status, internet services, technical support, and billing preferences can be evaluated to identify meaningful differences in retention.
6. **Billing information can support future financial analysis.** Monthly and total charges provide a foundation for estimating the financial consequences of customer attrition.

These insights are descriptive. They identify patterns in the available data but do not establish that any individual factor directly causes churn.

11. Strategic Recommendations

Recommendation 1: Improve Month-to-Month Customer Retention

Prioritize month-to-month customers because this group has the highest observed churn rate among the three contract types. Test flexible loyalty benefits, renewal incentives, and longer-term contract offers where appropriate.

Success metric: Reduce the month-to-month churn rate from its observed baseline of approximately 42.71%.

Recommendation 2: Investigate Electronic Check Churn

Review billing-related customer feedback and payment experiences for electronic check users. Test whether simpler payment options and clearer billing communication are associated with improved retention.

Success metric: Reduce the electronic-check customer churn rate from its observed baseline of approximately 45.29%.

Recommendation 3: Strengthen Early Customer Engagement

Use tenure-based segmentation to identify customers who may benefit from onboarding assistance, service education, or proactive support.

Success metric: Measure early-tenure churn before and after implementing the intervention.

Recommendation 4: Evaluate Service-Related Retention Opportunities

Compare churn rates for customers with and without services such as online security, online backup, device protection, and technical support. Use the results to identify possible improvements in service value and customer experience.

Success metric: Track churn rates by service category and assess whether targeted interventions improve retention.

Recommendation 5: Establish Regular Churn Reporting

Develop a recurring report or dashboard showing total customers, churned customers, overall churn rate, contract-specific churn, payment-specific churn, and tenure-based churn.

Success metric: Monitor churn rate over time and evaluate whether retention initiatives produce measurable improvements.

12. Conclusion

This project demonstrates how Python-based data analysis and visualization can convert customer data into actionable business insights. Through data cleaning, descriptive statistics, and multiple visualizations, I examined customer churn across demographics, tenure, contract types, payment methods, and service subscriptions.

The analysis identifies month-to-month contracts and electronic check payments as important areas for further investigation. Their observed churn rates are approximately 42.71% and 45.29%, respectively, compared with the overall customer churn rate of 26.54%.

These findings provide a foundation for more targeted retention strategies, better customer segmentation, and future revenue-impact analysis. By monitoring churn rates consistently and evaluating the results of targeted interventions, the business can make more informed decisions about customer retention and service improvement.